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Abstract

Terms of service of on-line platforms too often contain clauses that are potentially unfair to the consumer. We present an experimental study where machine learning is employed to automatically detect such potentially unfair clauses. Results show that the proposed system could provide a valuable tool for lawyers and consumers alike.

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Notes

See the Council Directive 93/13/EEC on Unfair Terms in Consumer Contracts, art. 3.1.

We remark that, from the point of view of natural language processing, we are handling a pure sentence classification task, as we detect full statements and not directly single clauses.

In particular, we selected the ToS offered by: 9gag.com, Academia.edu, Airbnb, Amazon, Atlas Solutions, Betterpoints, Booking.com, Crowdtangle, Deliveroo, Dropbox, Duolingo, eBay, Endomondo, Evernote, Facebook, Fitbit, Google, Headspace, Instagram, Linden Lab, LinkedIn, Masquerade, Microsoft, Moves-app, musically, Netflix, Nintendo, Oculus, Onavo, Pokemon GO, Rovio, Skype, Skyscanner, Snapchat, Spotify, Supercell, SyncMe, Tinder, TripAdvisor, TrueCaller, Twitter, Uber, Viber, Vimeo, Vivino, WhatsApp, World of Warcraft, Yahoo, YouTube and Zynga.

http://claudette.eui.eu/ToS.zip .

Segmentation into sentences was made using the Stanford CoreNLP suite (see Sect. 5).

In particular, we selected the ToS offered by: Alibaba, Badoo, Goodreads, Groupon, Mozilla, Ryanair, Shazam, Slack, Zalando UK, eDreams.

https://stanfordnlp.github.io/CoreNLP/ .

Sampling takes into account the class distribution in the training set.

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Keywords

- Machine learning
- Terms of service
- Potentially unfair clauses
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